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Bayesian Statistics and Data Analysis

Lecture 5

Måns Magnusson

Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University

- Monte Carlo recap
- Markov Chain Monte Carlo (MCMC)
 - Gibbs sampling
 - Metropolis-Hastings
- MCMC Diagnostics
 - Warm-up
 - Convergence
 - S_{eff} , MCSE, and autocorrelation
- Difficult geometries
- Additional material



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Oral exam + mini projects

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 - Additional material
- Oral exam evaluation (good and bad)?
 - Mini project proposals



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It's all about expectations

$$E_{p(\theta|y)}[f(\theta)] = \int f(\theta) p(\theta|y) d\theta,$$

$$\text{where } p(\theta|y) = \frac{p(y|\theta)p(\theta)}{\int p(y|\theta)p(\theta)d\theta}$$

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We can use the unnormalized posterior $q(\theta|y) = p(y|\theta)p(\theta)$, for example, in

- Monte Carlo methods which can sample from $p(\theta^{(s)}|y)$ using only $q(\theta^{(s)}|y)$

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)})$$

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 - Inverse CDF works mainly for 1D



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 - Laplace, Variational*, EP* (Ch 4, 13*, next course)



Markov chains

- A Markov chain is a sequence in which the next state depends on the current state, but not on earlier states

$$p(\theta^{(t)} \mid \theta^{(t-1)}, \dots, \theta^{(0)}) = p(\theta^{(t)} \mid \theta^{(t-1)}).$$

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- Total variation distance is

$$d_{TV}(P_t, \pi) = \sup_A |P_t(A) - \pi(A)|.$$

It is the largest difference between the probabilities that P_t and π assign to the same event A

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- Convergence in total variation means $d_{\text{TV}}(P_t, \pi) \rightarrow 0$

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- **Goal in MCMC**: construct T with $\pi(\theta) = p(\theta \mid y)$

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A stationary distribution: two-state example

Consider a chain with states 1 and 2 and transition matrix

$$T = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}.$$

Each row gives the probabilities of the next state.

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$$P_0 = (1, 0), \quad P_1 = (0.8, 0.2), \quad P_2 = (0.72, 0.28), \quad \dots$$

and P_t approaches π

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- Stationary means that applying the transition leaves the distribution unchanged; convergence means approaching it from another starting distribution

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- Produce draws $\theta^{(t)}$ given $\theta^{(t-1)}$ from a Markov chain, with **stationary distribution** $p(\theta|y)$



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 - + generic
 - + combine sequence of easier Monte Carlo draws to form a Markov chain



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 - + combine sequence of easier Monte Carlo draws to form a Markov chain
 - + converges (in total variation) to $p(\theta|y)$ as $t \rightarrow \infty$
 - + for well-behaved chains, a central limit theorem holds for Monte Carlo averages
 - draws are dependent
 - constructing an **efficient** Markov chain is not always easy



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- Random variables $\theta^{(1)}, \theta^{(2)}, \dots$ where $\theta^{(t)}$ depends only on the previous state $\theta^{(t-1)}$

$$p(\theta^{(t)} | \theta^{(1)}, \dots, \theta^{(t-1)}) = p(\theta^{(t)} | \theta^{(t-1)})$$



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- Choose a transition distribution so the **stationary distribution** of the Markov chain is $p(\theta|y)$



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Subsection 1

Gibbs sampling



- Alternate sampling from conditional distributions
- For two parameters, one complete sweep is

$$\theta_1^{(t)} \sim p(\theta_1 \mid \theta_2^{(t-1)}, y),$$

$$\theta_2^{(t)} \sim p(\theta_2 \mid \theta_1^{(t)}, y).$$

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- More generally, update $j = 1, \dots, J$ in order using

$$\theta_j^{(t)} \sim p\left(\theta_j \mid \theta_1^{(t)}, \dots, \theta_{j-1}^{(t)}, \theta_{j+1}^{(t-1)}, \dots, \theta_J^{(t-1)}, y\right).$$

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- Under standard conditions, the distribution of $\theta^{(t)}$ converges to $p(\theta \mid y)$ as $t \rightarrow \infty$
- Several parameters can be updated together in a block

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Example: Bivariate Normal using Gibbs

$$\theta | y \sim \mathcal{N}(\bar{y}, \frac{1}{n}\Sigma), \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

Here σ_1 and σ_2 are marginal standard deviations, ρ is the correlation, and Σ is known.

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Using **Gibbs sampling**:

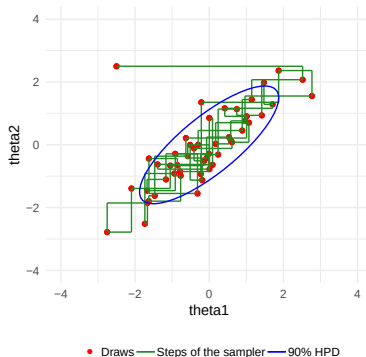
Given starting values $(\theta_1^{(0)}, \theta_2^{(0)})$, for $t = 1, 2, \dots$:

$$\theta_1^{(t)} \sim \mathcal{N}\left(\bar{y}_1 + \rho \frac{\sigma_1}{\sigma_2} (\theta_2^{(t-1)} - \bar{y}_2), \frac{\sigma_1^2}{n} (1 - \rho^2)\right),$$

$$\theta_2^{(t)} \sim \mathcal{N}\left(\bar{y}_2 + \rho \frac{\sigma_2}{\sigma_1} (\theta_1^{(t)} - \bar{y}_1), \frac{\sigma_2^2}{n} (1 - \rho^2)\right).$$



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demo



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- When the full conditional distributions have standard forms, exact Gibbs updates are easy to draw
 - Exact Gibbs updates have no proposal scale or rejection step
 - blocking, parameterization, and update order can still strongly affect efficiency



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- For a nonstandard full conditional, use a Metropolis–Hastings update within the Gibbs sweep



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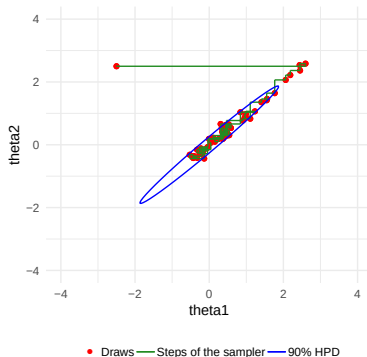


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- For a nonstandard full conditional, use a Metropolis–Hastings update within the Gibbs sweep
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- Slow if parameters are highly dependent in the posterior...



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Pair discussion: correlated Gibbs sampling

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- Both Gibbs samplers accept every update. Why does the sampler for the correlated target nevertheless produce fewer effective draws?
- What would you expect its trace plot and autocorrelation function to look like?



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Pair discussion: correlated Gibbs sampling

- Both Gibbs samplers accept every update. Why does the sampler for the correlated target nevertheless produce fewer effective draws?
- What would you expect its trace plot and autocorrelation function to look like?

After discussion:

- Each coordinate update moves only horizontally or vertically, while the posterior mass lies along a narrow diagonal region.
- The chain therefore moves slowly along that region, producing high autocorrelation and a small effective sample size.



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- How about sampling θ jointly?
 - e.g. it is easy to sample from multivariate normal



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 - e.g. it is easy to sample from multivariate normal
 - Can we use that to form a Markov chain?



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Subsection 2

Metropolis-Hastings



The Metropolis algorithm

- Algorithm

1. starting point $\theta^{(0)}$
2. $t = 1, 2, \dots$

(a) pick a proposal θ^* from a **proposal distribution** $J(\theta^*|\theta^{(t-1)})$.

(Proposal distribution has to be symmetric, i.e. $J(\theta_a|\theta_b) = J(\theta_b|\theta_a)$, for all θ_a, θ_b)

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$$r = \frac{p(\theta^*|y)}{p(\theta^{(t-1)}|y)}$$



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$$r = \frac{p(\theta^*|y)}{p(\theta^{(t-1)}|y)}$$

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$$\theta^{(t)} = \begin{cases} \theta^* & \text{with probability } \min(r, 1) \\ \theta^{(t-1)} & \text{otherwise} \end{cases}$$



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$$\theta^{(t)} = \begin{cases} \theta^* & \text{with probability } \min(r, 1) \\ \theta^{(t-1)} & \text{otherwise} \end{cases}$$

i.e., if $p(\theta^*|y) > p(\theta^{(t-1)}|y)$, always accept the proposal and otherwise accept the proposal with probability r



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- step c is executed by generating a random number from $\mathcal{U}(0, 1)$



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- rejection of a proposal increments the time t also by one i.e., the new state is the same as the previous state
- step c is executed by generating a random number from $\mathcal{U}(0, 1)$
- $p(\theta^*|y)$ and $p(\theta^{(t-1)}|y)$ have the same normalizing constant, so the unnormalized density $q(\cdot|y)$ can be used: **the constant cancels**



Rejection sampling vs. Metropolis rejection

- In rejection sampling, a rejected proposal is discarded
 - only accepted proposals are included in the output
 - the next proposal is drawn independently from g

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Rejection sampling vs. Metropolis rejection

- In rejection sampling, a rejected proposal is discarded
 - only accepted proposals are included in the output
 - the next proposal is drawn independently from g
- In the Metropolis algorithm, a rejected proposal is discarded, but the current state is repeated

$$\theta^{(t)} = \theta^{(t-1)}.$$

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- the repeated state is a valid draw from the Markov chain
- it must be retained when computing posterior summaries



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- In the Metropolis algorithm, a rejected proposal is discarded, but the current state is repeated

$$\theta^{(t)} = \theta^{(t-1)}.$$

- the repeated state is a valid draw from the Markov chain
 - it must be retained when computing posterior summaries
- Keeping only accepted Metropolis proposals changes the distribution and generally gives incorrect estimates



Consider a random-walk Metropolis proposal

$$\theta^* \sim \mathcal{N}(\theta^{(t-1)}, \sigma_p^2 I).$$

- Predict what happens when σ_p is much too small.
- Predict what happens when σ_p is much too large.
- For each case, what do you expect for the acceptance rate, trace plot, and autocorrelation?

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Proposal scale: discussion

Consider a random-walk Metropolis proposal

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- Predict what happens when σ_p is much too small.
- Predict what happens when σ_p is much too large.
- For each case, what do you expect for the acceptance rate, trace plot, and autocorrelation?

After discussion: Small proposals are often accepted but move slowly. Large proposals move farther when accepted, but are often rejected. Both choices can give high autocorrelation and low ESS.

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- Example: one bivariate observation $y = (y_1, y_2)$
 - $y \mid \theta \sim \mathcal{N}(\theta, \Sigma)$, with known covariance Σ and a flat prior $p(\theta) \propto 1$

$$\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} \Big| y \sim \mathcal{N} \left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \right)$$

- proposal distribution $J(\theta^* | \theta^{(t-1)}) = \mathcal{N}(\theta^* | \theta^{(t-1)}, \sigma_p^2 I)$

demo



Why the Metropolis algorithm works

- Intuitively, the chain spends more time in high-density regions
 - proposals to higher density are always accepted
 - proposals to lower density are accepted only sometimes

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- Convergence to that stationary distribution also requires a well-behaved chain
 - the proposal must allow the chain to reach all relevant regions of the target distribution
 - the chain must not be trapped in a deterministic cycle



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 - the proposal must allow the chain to reach all relevant regions of the target distribution
 - the chain must not be trapped in a deterministic cycle
- These ideas are formalized by **irreducibility** and **aperiodicity**
- They depend on both the target and the proposal; they do not follow merely because the target is a proper distribution



Detailed balance for the Metropolis algorithm

- For a symmetric proposal $J(\theta' | \theta)$, the Metropolis acceptance probability is

$$\alpha(\theta, \theta') = \min\left(1, \frac{p(\theta' | y)}{p(\theta | y)}\right).$$

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$$\begin{aligned} p(\theta | y) J(\theta' | \theta) \alpha(\theta, \theta') \\ = J(\theta' | \theta) \min(p(\theta | y), p(\theta' | y)). \end{aligned}$$

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- The self-transition caused by rejection balances trivially. Thus detailed balance holds for the full transition kernel T , and $p(\theta | y)$ is stationary



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- Generalization of Metropolis algorithm for **non-symmetric** proposal distributions
 - acceptance ratio includes ratio of proposal distributions

$$r = \frac{p(\theta^* | y) J(\theta^{(t-1)} | \theta^*)}{p(\theta^{(t-1)} | y) J(\theta^* | \theta^{(t-1)})}$$



- The ideal proposal is the target distribution itself
 - $J(\theta^*|\theta) \equiv p(\theta^*|y)$ gives independent draws with acceptance probability 1, but is not usually feasible

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- Proposal scale controls movement through the target
 - small scale: many accepted steps, but each step moves little
 - large scale: longer proposed steps, but many are rejected



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- [Interactive demonstration](#)



- Update one component while holding θ_{-j} fixed, proposing

$$\theta_j^* \sim p(\theta_j \mid \theta_{-j}, y).$$

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because each joint posterior factors into the same marginal for θ_{-j} times its full conditional

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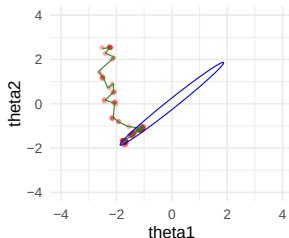
- Therefore every exact Gibbs proposal is accepted

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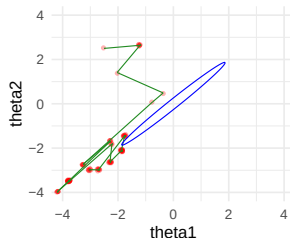


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- Usually does not scale well to high dimensions
 - if the shape does not match the whole distribution, efficiency drops



• Draws — Steps of the sampler — 90% HPI



• Draws — Steps of the sampler — 90% HPI



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Section 3

MCMC Diagnostics



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1. Do the chains show evidence of poor **convergence** or mixing?



What we want to know

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1. Do the chains show evidence of poor **convergence** or mixing?
2. How many **effective** samples do I have?



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Subsection 1

Warm-up



Warm-up

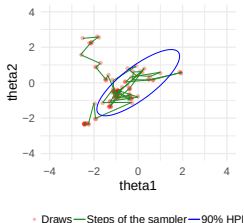
- Early draws can be strongly influenced by the starting value

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Warm-up

- Early draws can be strongly influenced by the starting value
 - the chain may not yet have reached the typical posterior region



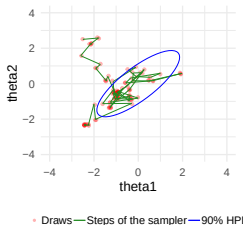
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Warm-up

- Early draws can be strongly influenced by the starting value
 - the chain may not yet have reached the typical posterior region

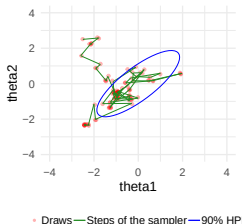


- Warm-up, also called **burn-in**, is the initial phase before retained sampling
 - do not use warm-up draws in posterior summaries
 - algorithms may adapt proposal or tuning parameters during warm-up



Warm-up

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 - the chain may not yet have reached the typical posterior region



- Warm-up, also called **burn-in**, is the initial phase before retained sampling
 - do not use warm-up draws in posterior summaries
 - algorithms may adapt proposal or tuning parameters during warm-up
- Discarding warm-up does not guarantee good sampling
 - the retained chains can still mix slowly or miss posterior regions
 - convergence and efficiency diagnostics are still required



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Subsection 2

Convergence



Assessing convergence

- Several Markov chains make convergence diagnostics easier

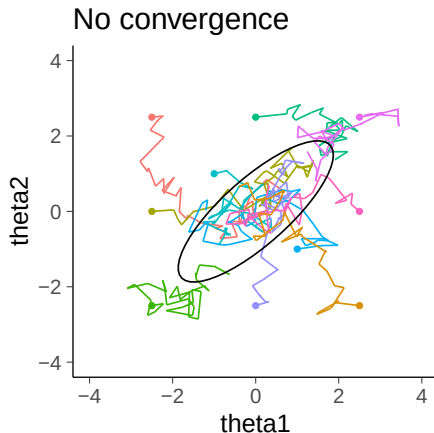
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Assessing convergence

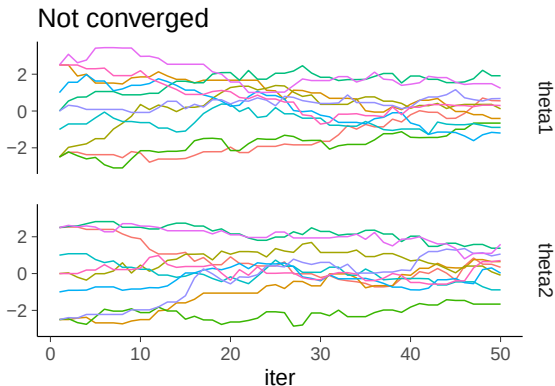
- Several Markov chains make convergence diagnostics easier
- Start chains from different, preferably overdispersed, starting points



- Remove warm-up draws and run chains long enough

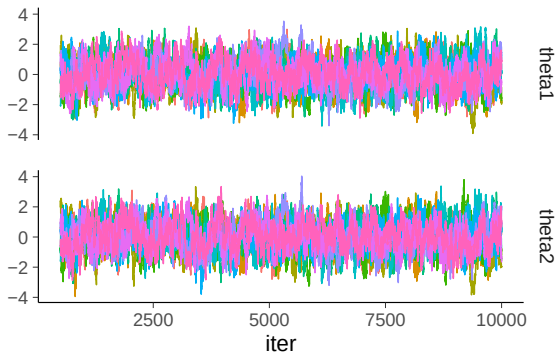


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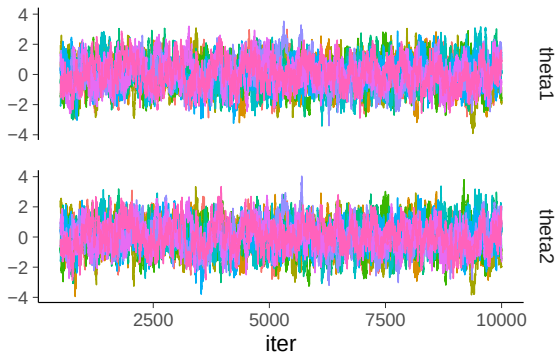
Visually converged



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Visually converged



Visual convergence check is not sufficient

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$\hat{R}$: comparison of within and between variances of the chains

- $\hat{R}$ or potential scale reduction factor (PSRF)
- Compare means and variances of the chains

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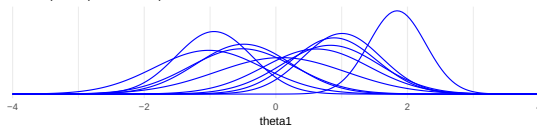


$\hat{R}$: comparison of within and between variances of the chains

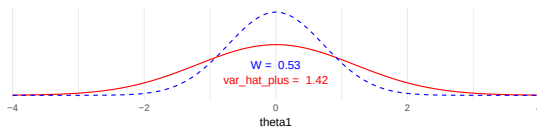
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- $\hat{R}$ or **potential scale reduction factor** (PSRF)
- Compare means and variances of the chains
 W = within-chain variance estimate
 $\widehat{\text{var}}^+$ = total variance estimate

50 warmup, 50 post warmup iterations



Rhat = 1.64



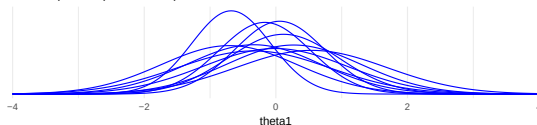


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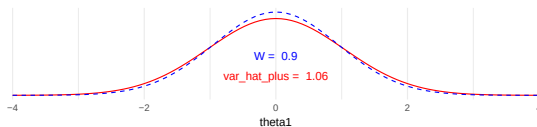
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- $\hat{R}$ or **potential scale reduction factor** (PSRF)
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500 warmup, 500 post warmup iterations



Rhat = 1.08



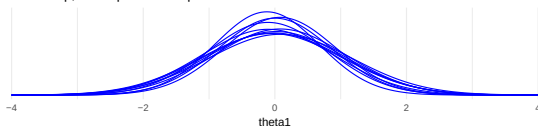


$\hat{R}$: comparison of within and between variances of the chains

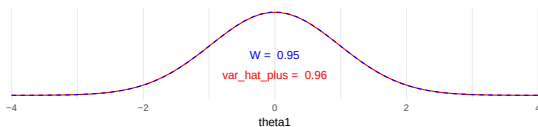
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 W = within-chain variance estimate
 $\widehat{\text{var}}^+$ = total variance estimate

5000 warmup, 5000 post warmup iterations



Rhat = 1





Classical $\hat{R}$: interpretation

- The classical diagnostic compares a total variance estimate $\widehat{\text{var}}^+$ with the within-chain variance W

$$\hat{R} = \sqrt{\frac{\widehat{\text{var}}^+}{W}}.$$

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- $\hat{R}$ close to 1 means the between-chain and within-chain variation are similar
- $\hat{R} > 1$ indicates that the chains disagree more than expected from their within-chain variation

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- $\hat{R}$ close to 1 means the between-chain and within-chain variation are similar
- $\hat{R} > 1$ indicates that the chains disagree more than expected from their within-chain variation
- A value close to 1 is necessary but not sufficient: also inspect trace plots, ESS, and MCSE



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Subsection 3

S_{eff} , MCSE, and autocorrelation



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- Notation: M chains, N retained draws per chain, and $S = MN$ retained draws in total
- For well-behaved chains, Monte Carlo averages converge to posterior expectations

$$E_{p(\theta|y)}[f(\theta)] \approx \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)})$$

- A Markov chain central limit theorem can then be used to estimate Monte Carlo error, but dependence makes this more difficult
 - evaluation of *effective* sample size, S_{eff} .

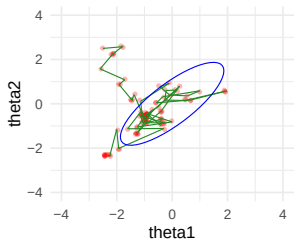


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- The autocorrelation function measures dependence between draws separated by a given lag
 - high autocorrelation means that successive draws contain less new information
 - it helps compare the efficiency of MCMC methods



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Autocorrelation: one worked example

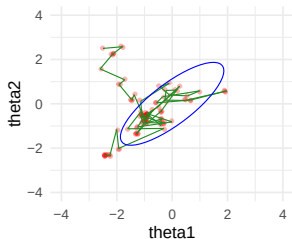


• Draws — Steps of the sampler — 90% HPD

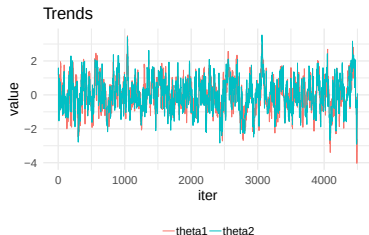


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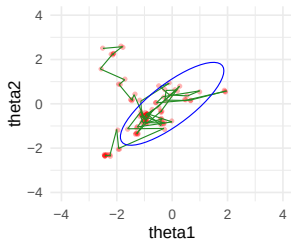
— θ_1 — θ_2



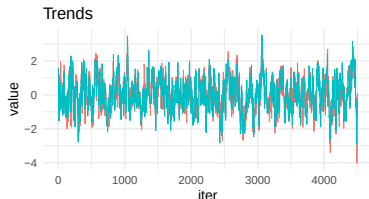
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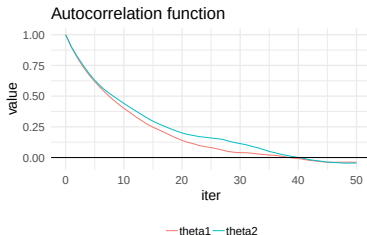
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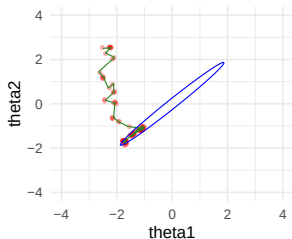


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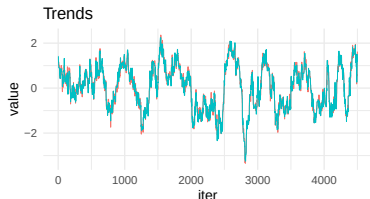


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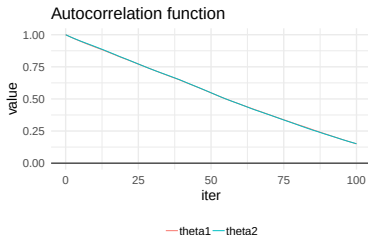
Autocorrelation: comparison example 1



• Draws — Steps of the sampler — 90% HPI



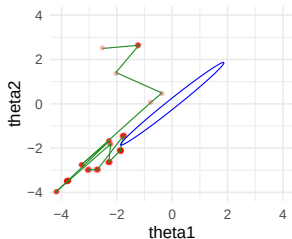
— θ_1 — θ_2



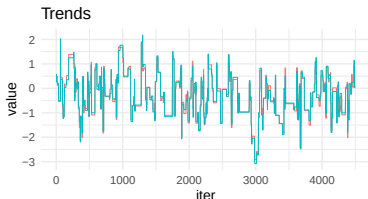


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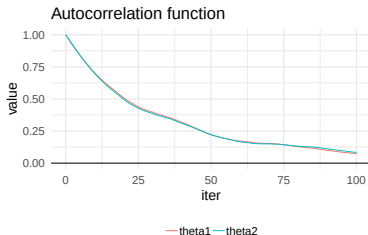
Autocorrelation: comparison example 2



• Draws — Steps of the sampler — 90% HPD



— θ_1 — θ_2



— θ_1 — θ_2



Effective sample size and MCSE

- Suppose we estimate $\mu_f = E[f(\theta) \mid y]$ using

$$\hat{\mu}_f = \frac{1}{S} \sum_{s=1}^S f(\theta^{(s)}).$$

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- Larger effective sample size means a more precise Monte Carlo estimate; doubling ESS reduces MCSE by a factor of $\sqrt{2}$

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- For lag k , estimate the expected squared change within chains

$$\hat{V}_k = \frac{1}{M(N-k)} \sum_{m=1}^M \sum_{n=1}^{N-k} (\theta_{n+k,m} - \theta_{n,m})^2.$$

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- Small squared changes imply high autocorrelation; large changes imply lower autocorrelation
- Using $\widehat{\text{var}}^+$ incorporates information from all chains and penalizes chains that have not mixed well

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Integrated autocorrelation time

- Let ρ_k be the autocorrelation at lag k . The integrated autocorrelation time is

$$\tau = 1 + 2 \sum_{k=1}^{\infty} \rho_k, \quad S_{\text{eff}} = \frac{S}{\tau}.$$

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- τ measures how many correlated draws provide roughly the same information as one independent draw
- Estimated autocorrelations become noisy at long lags
 - practical estimators truncate the sum at a finite lag K

$$\hat{\tau} = 1 + 2 \sum_{k=1}^K \hat{\rho}_k$$

- statistical software chooses the maximum lag K adaptively

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Effective sample size, S_{eff}

Fastest ACF decay: $\hat{\tau} \approx 24$ and $\hat{S}_{\text{eff}} \approx 188$ from $S = 4500$
retained draws

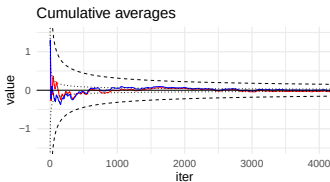
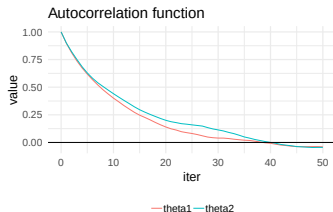
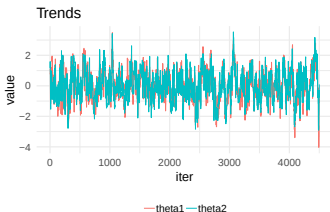
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Effective sample size, S_{eff}

Fastest ACF decay: $\hat{\tau} \approx 24$ and $\hat{S}_{\text{eff}} \approx 188$ from $S = 4500$ retained draws



— theta1 — theta2 — 95% interval for MCMC error — 95% interval for indepen

$$\hat{\tau} = 1 + 2 \sum_{k=1}^K \hat{\rho}_k$$

$$\approx 24$$

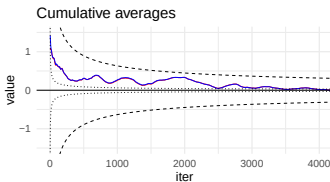
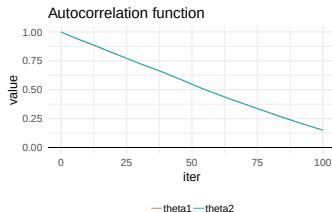
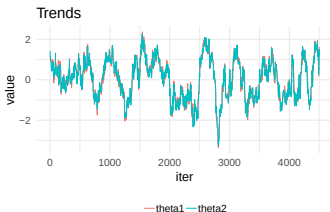
$$\hat{S}_{\text{eff}} = \frac{4500}{24} \approx 188$$



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Effective sample size, S_{eff}

Slowest ACF decay: $\hat{\tau} \approx 104$ and $\hat{S}_{\text{eff}} \approx 43$ from $S = 4500$ retained draws



— theta1 — theta2 — 95% interval for MCMC error ... 95% interval for indepen

$$\hat{\tau} = 1 + 2 \sum_{k=1}^K \hat{\rho}_k$$

$$\approx 104$$

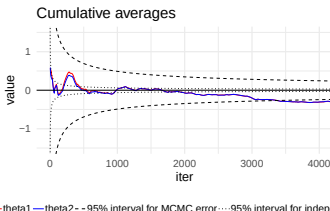
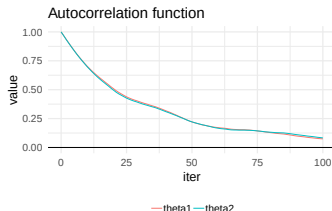
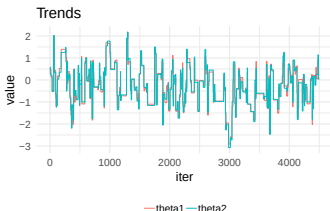
$$\hat{S}_{\text{eff}} = \frac{4500}{104} \approx 43$$



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Effective sample size, S_{eff}

Across all three examples: slower ACF decay $\Rightarrow$ larger $\hat{\tau} \Rightarrow$ smaller $\hat{S}_{\text{eff}}$.



$$\hat{\tau} = 1 + 2 \sum_{k=1}^K \hat{\rho}_k$$

$$\approx 63$$

$$\hat{S}_{\text{eff}} = \frac{4500}{63} \approx 71$$



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Section 4

Difficult geometries



Difficult geometry: changing scale and direction

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- **Nonlinear dependence:** posterior mass follows a curved, narrow region
 - a fixed random-walk direction repeatedly proposes points outside the high-density region
 - symptoms: slow movement, high autocorrelation, and low ESS



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Difficult geometry: changing scale and direction

- **Nonlinear dependence:** posterior mass follows a curved, narrow region
 - a fixed random-walk direction repeatedly proposes points outside the high-density region
 - symptoms: slow movement, high autocorrelation, and low ESS
- **Funnel geometry:** the appropriate step size changes across the posterior
 - a step size suitable for the wide part is too large for the neck
 - a step size suitable for the neck moves very slowly in the wide part
 - symptoms: chains explore different scales or rarely enter the neck

Banana demo

Funnel demo



- **Multimodality:** chains can remain in different modes
 - chain means disagree and $\hat{R}$ can remain above 1
 - if every chain starts in the same mode, diagnostics may fail to reveal the missing mode

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- **Multimodality:** chains can remain in different modes
 - chain means disagree and $\hat{R}$ can remain above 1
 - if every chain starts in the same mode, diagnostics may fail to reveal the missing mode
- **Non-identifiability:** many parameter combinations fit the data almost equally well
 - the posterior contains a ridge rather than one compact region
 - symptoms: drifting traces, strong dependence, and low ESS

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Difficult geometry: hidden regions

- **Multimodality**: chains can remain in different modes
 - chain means disagree and $\hat{R}$ can remain above 1
 - if every chain starts in the same mode, diagnostics may fail to reveal the missing mode
- **Non-identifiability**: many parameter combinations fit the data almost equally well
 - the posterior contains a ridge rather than one compact region
 - symptoms: drifting traces, strong dependence, and low ESS
- **Heavy tails**: occasional extreme draws dominate summaries
 - means or variances may not exist, so the usual CLT and MCSE may not apply

Multimodal demo

Ridge demo



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MCMC diagnostic workflow

1. Run several chains from different, reasonably dispersed starting values
2. Remove warm-up draws before computing posterior summaries
3. Inspect trace plots: chains should explore the same regions without persistent trends
4. Check that $\hat{R}$ is close to 1 for all reported quantities
5. Check ESS and MCSE for the posterior summaries that matter
6. If diagnostics are poor, investigate the proposal, parameterization, model, and posterior geometry



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Running longer helps only if the chain is exploring the relevant posterior regions.



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Section 5

Additional material



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 - **Additional material**
- BUGS popularized automatic Gibbs sampling for Bayesian models
 - WinBUGS and OpenBUGS provided widely used BUGS implementations
 - JAGS follows a similar model-description and sampling approach
 - These systems work especially naturally for models with conditionally conjugate components



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$\hat{R}$

- M chains, each having N draws

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- M chains, each having N draws
- Within chains variance W

$$W = \frac{1}{M} \sum_{m=1}^M s_m^2, \text{ where } s_m^2 = \frac{1}{N-1} \sum_{n=1}^N (\theta_{nm} - \bar{\theta}_{.m})^2$$



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- Between chains variance B

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$$\text{where } \bar{\theta}_{.m} = \frac{1}{N} \sum_{n=1}^N \theta_{nm}, \bar{\theta}_{..} = \frac{1}{M} \sum_{m=1}^M \bar{\theta}_{.m}$$



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- B/N is variance of the means of the chains
- Estimate total variance $\text{var}(\theta|y)$ as a weighted mean of W and B

$$\widehat{\text{var}}^+(\theta|y) = \frac{N-1}{N} W + \frac{1}{N} B$$



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- Before chains have fully explored the posterior, W typically *underestimates* the marginal posterior variance
 - single chains have not yet visited all points in the distribution
 - when $N \rightarrow \infty$, $E(W) \rightarrow \text{var}(\theta|y)$



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- Before chains have fully explored the posterior, W typically *underestimates* the marginal posterior variance
 - single chains have not yet visited all points in the distribution
 - when $N \rightarrow \infty$, $E(W) \rightarrow \text{var}(\theta|y)$
- Comparing these two estimates motivates

$$\hat{R} = \sqrt{\frac{\widehat{\text{var}}^+}{W}}$$



Geyer's adaptive window estimator of τ

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- The maximum lag K can be chosen adaptively
 - for a stationary, reversible Markov chain
 - let $\Gamma_m = \rho_{2m} + \rho_{2m+1}$, the sum of two consecutive autocorrelations
 - Γ_m is a positive, decreasing, and convex function of m



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Geyer's adaptive window estimator of τ

- The maximum lag K can be chosen adaptively
 - for a stationary, reversible Markov chain
 - let $\Gamma_m = \rho_{2m} + \rho_{2m+1}$, the sum of two consecutive autocorrelations
 - Γ_m is a positive, decreasing, and convex function of m
- Initial positive sequence estimator (Geyer's IPSE)
 - Choose the largest m such that all values of the sequence $\hat{\Gamma}_1, \dots, \hat{\Gamma}_m$ are positive

